

Hungry Student — College Food Delivery App

Java OOP Mini-Project — Assignment Handout

Build a small Java program for a college food delivery app. A student can order food from a restaurant, make a payment, and get the food delivered.

Classes to Create

1. Student

Variables:

```
name
studentId
walletBalance
hungerLevel
```

Methods:

```
addMoney()
orderFood()
showBalance()
```

Funny rules:

- If **hungerLevel** is greater than 80 → print *“EMERGENCY! Feed the student immediately!”*
- The student cannot spend more than **walletBalance**.

2. FoodOrder

Variables:

```
orderId
foodName
price
quantity
```

Methods:

```
calculateTotal()
showOrder()
```

Funny rules:

- If food is **“Biryani”** → print *“Student has chosen happiness”*
- If **quantity > 3** → print *“Bro, are you feeding the entire class?”*

3. Payment

Create different payment options:

```
UPI
```

```
CreditCard  
Cash
```

Each payment method should have:

```
pay()
```

4. Delivery

Create different delivery methods:

```
BikeDelivery  
DroneDelivery
```

Each should have:

```
deliver()
```

Funny rules:

- Bike → *“Your food is coming by bike”*
- Drone → *“Your food is flying to you”*

Your Task

Your program must demonstrate all 4 OOP pillars:

```
Encapsulation  
Abstraction  
Inheritance  
Polymorphism
```

Do not create everything in one class.

The student should be able to run something like:

```
Student: Hemanth  
Food: Biryani  
Quantity: 2  
Total: Rs.300  
  
Payment: UPI  
Payment successful!  
  
Delivery: Bike  
Your biryani is on the way
```

Challenge

Students decide themselves where and how to apply each OOP pillar. There is more than one correct design — be ready to explain **why** you structured your classes the way you did.

Submission: source code (.java files) that compiles and runs, plus a short note on where each OOP pillar was applied.